



CAPSTONE PROJECT

HOUSEHOLD ENERGY OPTIMIZATION SYSTEM

PRESENTED BY

STUDENT NAME: LALIT KUMAR

COLLEGE NAME: KUCHAMAN COLLEGE/MDSU

DEPARTMENT: BCA

EMAIL ID: lalitrnwa2216@gmail.com



OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

- Efficient energy management is a major challenge in modern households due to increasing electricity demand and fluctuating consumption patterns. Traditional energy monitoring systems lack predictive capabilities and can't forecast future energy consumption accurately.
- Unpredictable household energy usage leads to inefficient resource utilization, higher electricity bills, and difficulty in implementing smart grid solutions.
- Therefore, there is a need for an intelligent system capable of predicting household energy consumption using historical data and machine learning techniques.

PROPOSED SOLUTION:

- The proposed system aims to develop an AI/ML-based regression model to predict household energy consumption using historical power usage data. **The solution will consist of the following components:**
- **Data Collection:**
 - Historical electricity consumption data was collected from the UCI Machine Learning Repository – Individual Household Electric Power Consumption Dataset, which is also available on Kaggle.
- **Data Preprocessing:**
 - The collected data is cleaned and preprocessed to improve model accuracy.
 - **This includes:** Handling missing values (represented as '?'), Converting string values into numeric format
Removing inconsistencies and outliers, Feature engineering (e.g., extracting hour from Date-Time for time-based analysis)
- **Machine Learning Algorithm:**
 - Implement regression algorithms such as:
 - Linear Regression
 - Random Forest Regressor
 - These models are used to predict future household energy consumption based on historical data patterns.
- **Model Evaluation:**
 - The model performance is evaluated using appropriate metrics such as:
 - Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
 - The model is fine-tuned based on evaluation results and continuous monitoring to improve prediction accuracy.

SYSTEM APPROACH:

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the Household Energy Consumption Prediction System.

- **System Requirements**

Python 3.x , Minimum 4GB RAM, Windows / Linux Operating System

- **Technology Used**

Python, VS Code (Development Environment), GitHub (Version Control)

- **Libraries Used**

pandas (Data manipulation), numpy (Numerical operations),
matplotlib (Data visualization), scikit-learn (Machine Learning algorithms)

- **Dataset**

UCI Machine Learning Repository
Individual Household Electric Power Consumption Dataset

- **Algorithm Selection:**

- Linear Regression was used as a baseline model to capture linear relationships, while Random Forest Regressor was selected for its higher accuracy, ability to model complex non-linear patterns, reduced overfitting, and strong performance on large datasets.

- **Data Input:**

- The model uses historical household electrical parameters such as Global_reactive_power, Voltage, Global_intensity, Sub_metering_1, Sub_metering_2, Sub_metering_3, and time-based features (hour/date) to predict Global_active_power (total energy consumption).

- **Training Process:**

- The dataset was split into 80% training and 20% testing data, the models were trained on the training set with hyperparameter tuning ($n_estimators = 100$), and the system learned patterns between electrical parameters and energy consumption.

- **Prediction Process:**

- The trained model processes input electrical features through regression algorithms to generate predicted energy consumption values, which are then evaluated against actual values using MAE and RMSE metrics.

RESULT:

Present the results of the machine learning model in terms of its accuracy and effectiveness in predicting household energy consumption.

Include visualizations and comparisons between predicted and actual energy values to highlight the model's performance.

Model Evaluation:

<u>Model</u>	<u>MAE</u>	<u>RMSE</u>
Linear Regression	0.0364	0.0601
Random Forest	0.0176	0.0375

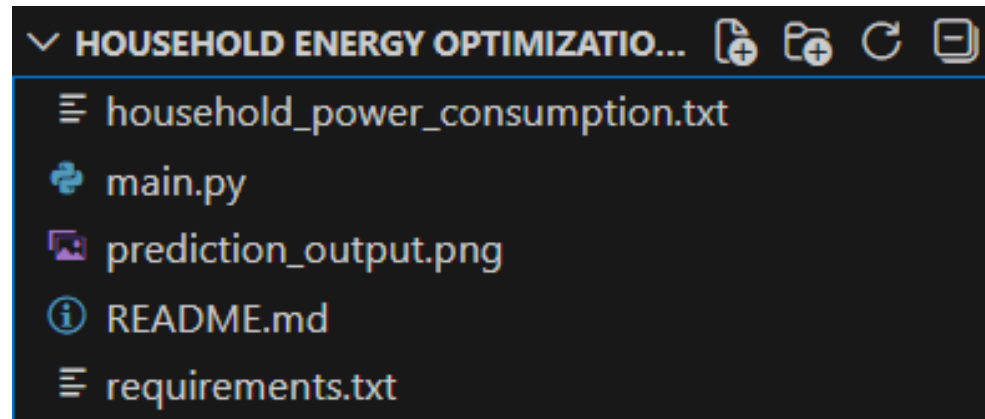
Observations:

- Random Forest achieved lower error values compared to Linear Regression.
- Strong overlap between actual and predicted energy consumption values.
- The model demonstrates high prediction accuracy and reliability.

Household Electric Power Consumption Dataset :

```
Date;Time;Global_active_power;Global_reactive_power;Voltage;Global_intensity;Sub_metering_1;Sub_metering_2;Sub_metering_3
16/12/2006;17:24:00;4.216;0.418;234.840;18.400;0.000;1.000;17.000
16/12/2006;17:25:00;5.360;0.436;233.630;23.000;0.000;1.000;16.000
16/12/2006;17:26:00;5.374;0.498;233.290;23.000;0.000;2.000;17.000
16/12/2006;17:27:00;5.388;0.502;233.740;23.000;0.000;1.000;17.000
16/12/2006;17:28:00;3.666;0.528;235.680;15.800;0.000;1.000;17.000
16/12/2006;17:29:00;3.520;0.522;235.020;15.000;0.000;2.000;17.000
16/12/2006;17:30:00;3.702;0.520;235.090;15.800;0.000;1.000;17.000
16/12/2006;17:31:00;3.700;0.520;235.220;15.800;0.000;1.000;17.000
16/12/2006;17:32:00;3.668;0.510;233.990;15.800;0.000;1.000;17.000
```

Project File Structure :



Final Output Prediction

```
PS D:\AICTE Project\Household Energy Optimization System Project> python main.py
>>
Loading Dataset...

First 5 Rows:
   Date      Time Global_active_power Global_reactive_power Voltage Global_intensity Sub_metering_1 Sub_metering_2 Sub_metering_3
0 16/12/2006 17:24:00          4.216           0.418  234.840           18.400           0.000           1.000           17.0
1 16/12/2006 17:25:00          5.360           0.436  233.630           23.000           0.000           1.000           16.0
2 16/12/2006 17:26:00          5.374           0.498  233.290           23.000           0.000           2.000           17.0
3 16/12/2006 17:27:00          5.388           0.502  233.740           23.000           0.000           1.000           17.0
4 16/12/2006 17:28:00          3.666           0.528  235.680           15.800           0.000           1.000           17.0

Cleaning Data...

Data Types After Cleaning:
Date      datetime64[us]
Time      str
Global_active_power    float64
Global_reactive_power  float64
Voltage              float64
Global_intensity      float64
Sub_metering_1        float64
Sub_metering_2        float64
Sub_metering_3        float64
Hour                int32
dtype: object

Training Linear Regression...
Training Random Forest...

Model Evaluation Results:

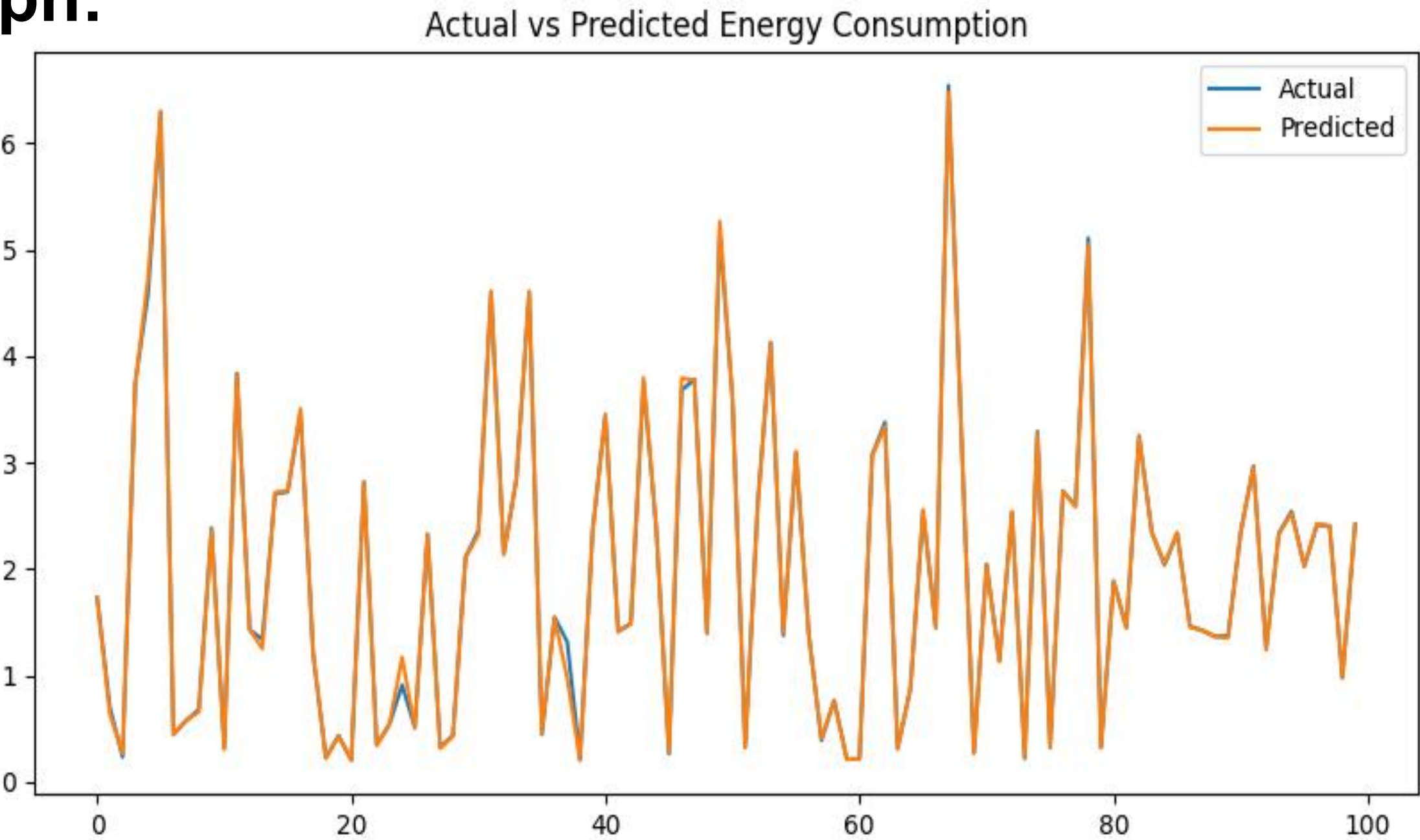
Linear Regression:
MAE: 0.036455968663049586
RMSE: 0.060140607306029034

Random Forest:
MAE: 0.017629102719002063
RMSE: 0.03759196800656804

Graph saved as prediction_output.png

Project Completed Successfully!
```

Graph:



CONCLUSION:

- The Household Energy Optimization System successfully predicts energy consumption using machine learning algorithms.
- Random Forest outperformed Linear Regression in terms of accuracy and error reduction.
- **The system can assist in:**
 - Smart energy planning
 - Reducing electricity wastage
 - Supporting intelligent grid systems
 - Improving household energy efficiency

FUTURE SCOPE:

The system can be enhanced by incorporating real-time energy data, improving model performance, and integrating with smart home and smart grid technologies, along with future expansions such as IoT-based smart meter integration, real-time energy monitoring dashboards, cloud deployment using Azure ML, deep learning models (LSTM) for time-series forecasting, mobile applications for energy tracking, and smart grid integration to improve scalability and efficiency.

REFERENCES:

- List and cite relevant sources, research papers, and articles that were instrumental in developing the proposed solution, including academic papers on energy consumption forecasting, machine learning algorithms, and best practices in data preprocessing and model evaluation.
- UCI Machine Learning Repository – [Household Electric Power Consumption Dataset](#)
- Research papers on Energy Consumption Forecasting
- Documentation of scikit-learn (Machine Learning Library)
- Python Official Documentation

GitHub Link: [Link](#)

Thank You